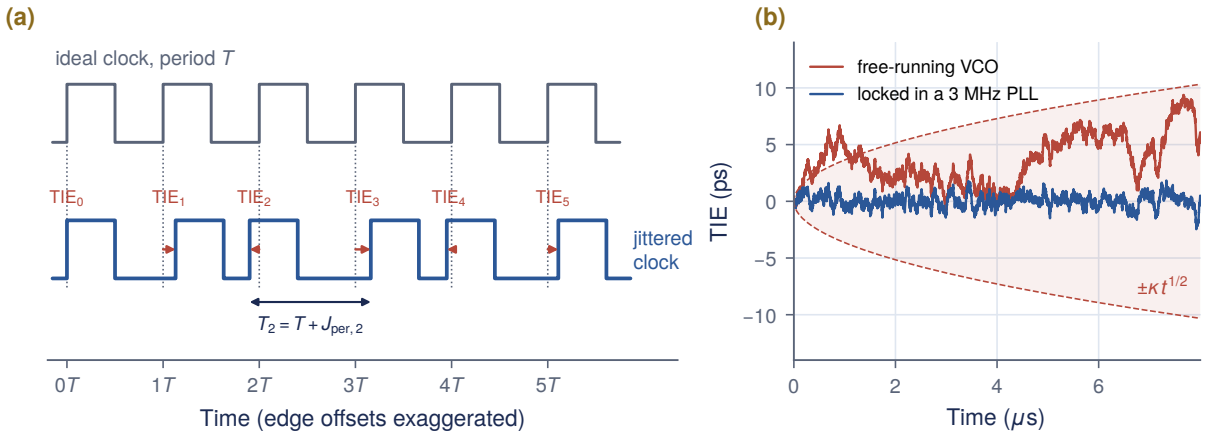

(a)
ideal clock, period T
TIE0
TIE1
TIE2
TIE3
TIE4
TIE5
jittered clock
T2 = T + Jper, 2
0T
1T
2T
3T
4T
5T
Time (edge offsets exaggerated)
(b)
free-running VCO
locked in a 3 MHz PLL
TIE (ps)
10
5
0
−5
−10
±κt1/2
0
2
4
6
Time (μs)